Assignment 3: Transformer is All You Need

Tiny Shakespeare Next-Token Model (From-Scratch Transformer)

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Code: https://github.com/zz3231/COMS4995-Applied-Machine-Learning-HW/tree/main/HW3

1 Introduction

This report describes a **minimal causal Transformer** language model implemented in PyTorch for **next-token prediction** on the Tiny Shakespeare corpus. The training objective is standard causal LM: maximize $\log p(x_t \mid x_{<t})$ at each position using cross-entropy. We report **validation perplexity** $\mathrm{PPL} = \exp(\mathcal{L})$, where $\mathcal{L}$ is mean validation cross-entropy in nats; lower is better.

The model uses two Transformer blocks, $d_{\text{model}} = 128$, eight heads, feed-forward width 512, **RMSNorm**, **sinusoidal positional encodings**, and **causal** multi-head self-attention. Tokenization is **Byte-Pair Encoding** with vocabulary size capped at **500**, fitted **only on the training split** after an 80/20 character-level partition so validation tokens do not influence merge statistics.

2 Methods

2.1 Data and tokenization

Tiny Shakespeare is loaded from the usual Karpathy URL (∼1.1M characters). The first 80% of characters are `train_text`; the rest is `val_text`. A ByteLevel BPE tokenizer (`tokenizers`) is trained on `train_text` with `vocab_size`=500 and `min_frequency`=2. Overlapping windows of **65** tokens with stride **32** yield ∼12.8k training windows; each window yields **64** supervised positions (input = positions $0{:}T-1$, target = positions $1{:}T$).

2.2 Model architecture

We use **pre-LN** blocks: $x \leftarrow x + \mathrm{Dropout}(\mathrm{Sublayer}(\mathrm{RMSNorm}(x)))$ for attention and FFN sublayers. **Causal self-attention** applies a lower-triangular mask before softmax so query i cannot attend to keys $j > i$. **Sinusoidal** positional vectors are added to token embeddings (scaled by $\sqrt{d_{\text{model}}}$). A final RMSNorm and linear **language-model head** map hidden states to vocabulary logits. Total parameter count is on the order of one million, small enough for a laptop CPU or a single Colab GPU.

2.3 Training and evaluation

Optimization uses **AdamW** (learning rate 3×10^{-4}, weight decay 0.01), batch size **32**, dropout 0.1, and **gradient clipping** (global norm 1.0). The main model trains for **25** epochs. Random seeds are fixed; validation uses `torch.no_grad()`. Metrics are mean cross-entropy and PPL = exp(CE) on the validation loader.

2.4 Extra experiments

(A) Positional encoding ablation: the same architecture without sinusoidal PE is trained for **15** epochs. Figure 4 plots validation PPL for the first 15 epochs of the main run versus all 15 epochs of the no-PE model (aligned x-axis, fair epoch-matched comparison).

(B) Shorter context: datasets use block size **33** and stride **16** (32 prediction positions). A fresh model trains for **12** epochs. This run is **not** directly comparable to the main model (different sequence length

T, different number of epochs, and a separate weight initialization); it illustrates the context-length versus compute tradeoff.

3 Results

3.1 Training curves and perplexity

Figure 1 shows training and validation cross-entropy and validation perplexity. Training CE decreases smoothly to ≈ 3.38 by epoch 25; validation CE ends at **3.7217**, giving PPL = **41.33**.

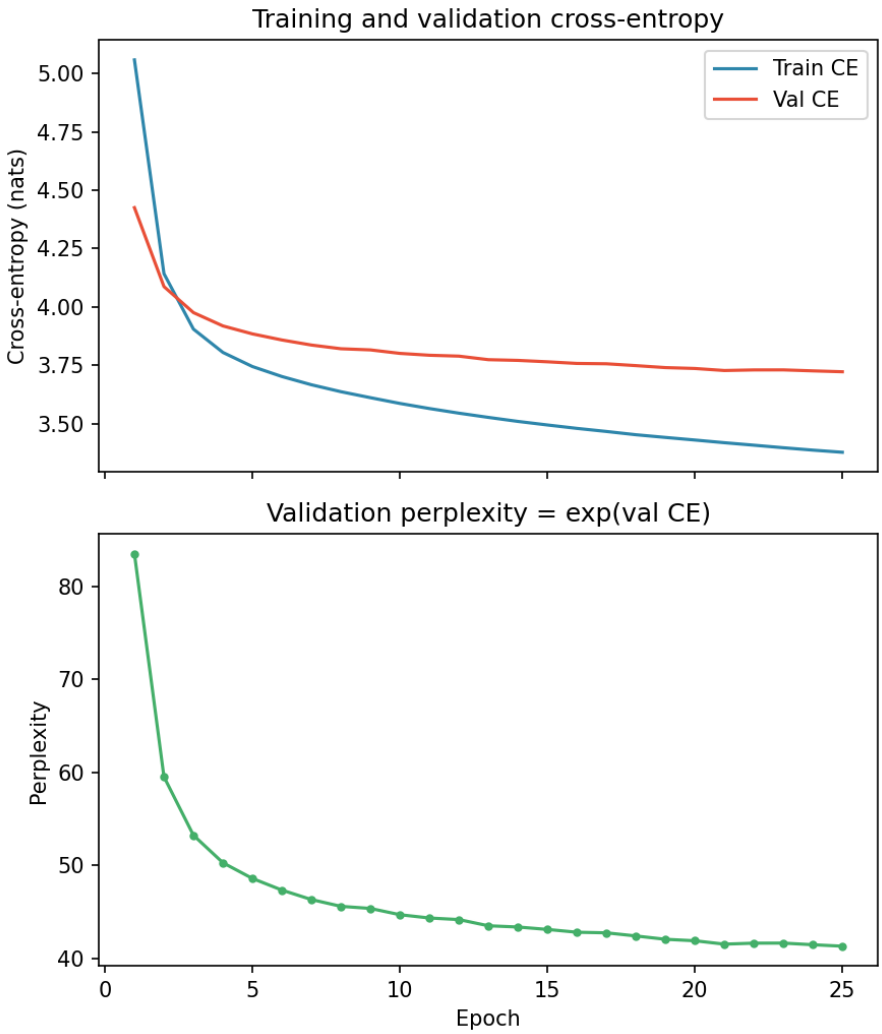

Figure 1: Training/validation cross-entropy (top) and validation perplexity (bottom) over 25 epochs.

Table 1: Hyperparameters and reported metrics (from the submitted notebook run).

Setting	Value
Vocab	500 BPE (train-only)
Block / stride (main)	65 / 32 (64 positions)
Batch size	32
d_{model}, layers, heads, FFN	128, 2, 8, 512
Optimizer	AdamW, lr=3×10^{-4}, wd=0.01
Gradient clipping	max norm 1.0
Main training	25 epochs
Final train CE / val CE / val PPL	3.38 / 3.72 / **41.33**
No-PE ablation	15 epochs; final val PPL **43.58**
Short context	block 33, 12 epochs; final val PPL **40.19**

3.2 Attention heatmaps

Heatmaps use one **validation** prefix (decoded ByteLevel subwords on both axes). The mask enforces a **strictly lower-triangular** pattern: each row (query position) has nonzero mass only on columns $\leq$ that index.

Layer 0, head 0 (Figure 2). The early-layer map is relatively **diffuse** along allowed past positions: several predecessors receive moderate weight rather than a single sharp peak. Some mass concentrates on the most recent tokens (a local "recency" bias), which is typical when the model begins combining raw subword identities before higher-level structure is explicit.

Layer 1, head 2 (Figure 3). The deeper head shows **sparser**, more peaked assignments on a subset of prior subwords. Different heads can specialize (e.g., attending to separators, frequent fragments, or specific predecessors); comparing L0H0 and L1H2 on the **same** sequence makes head diversity visible without conflating sample noise.

We did **not** re-render attention at early training epochs; all heatmaps are from the **converged** main model, so we cannot claim time-lapse evolution of attention within this report.

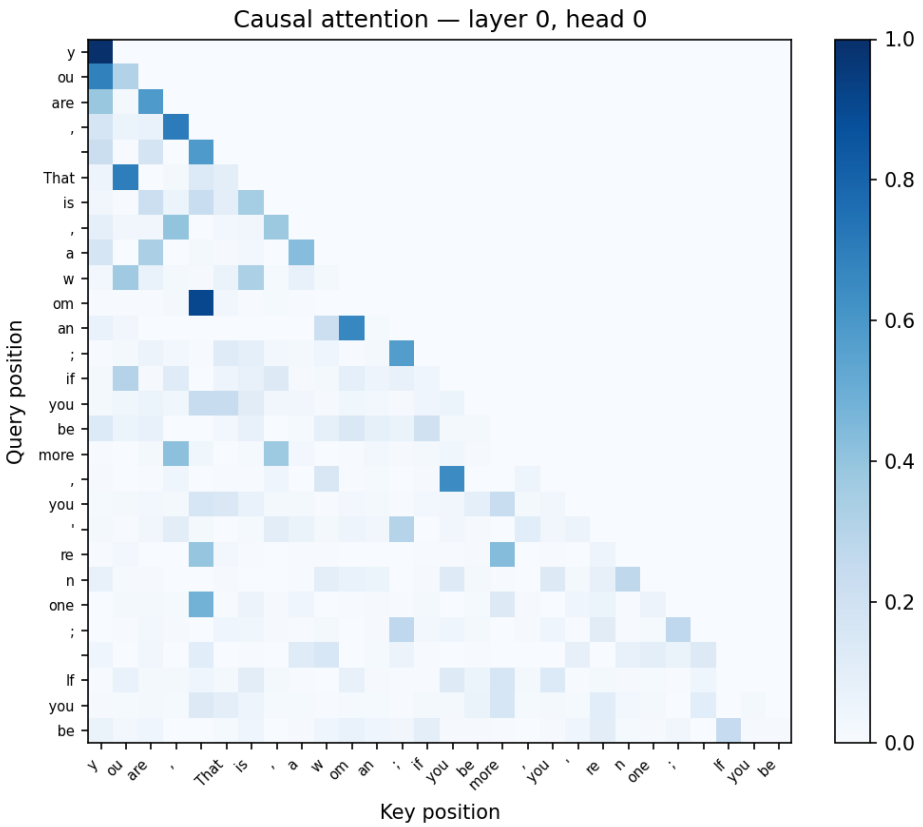

Figure 2: Layer 0, head 0: causal attention on a fixed validation prefix (subword labels).

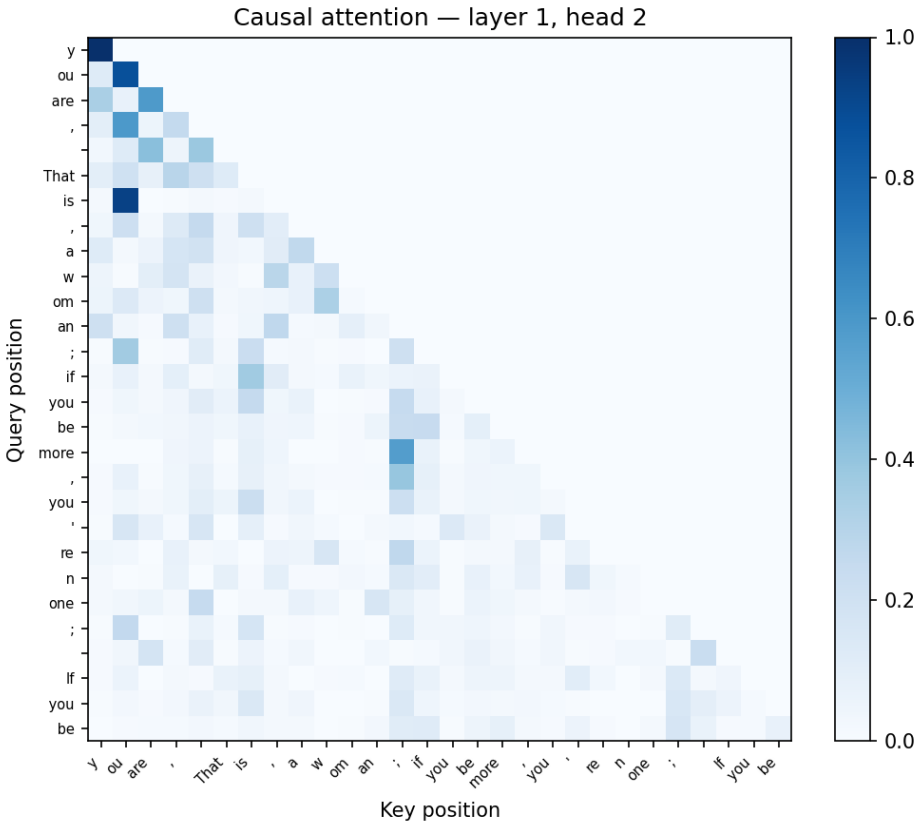

Figure 3: Layer 1, head 2: same prefix as Figure 2; sparser peaks than the early-layer head.

3.3 Ablations and context length

Figure 4 shows that removing sinusoidal PE **raises** validation PPL at the same epoch index: after 15 epochs, no-PE reaches **43.58** versus **43.13** for the main model at epoch 15 (and **41.33** after the full 25-epoch main run). The gap supports the view that absolute position cues help even on this small model and corpus.

Figure 5 gives validation PPL for the short-context setup. The final value **40.19** is **not** a claim that shorter context beats the main model overall—training length and T differ. The intended takeaway is **operational**: shorter T reduces attention memory and FLOPs (both scale roughly as $O(T^2)$ in this implementation) at the cost of less context for long-range dependencies.

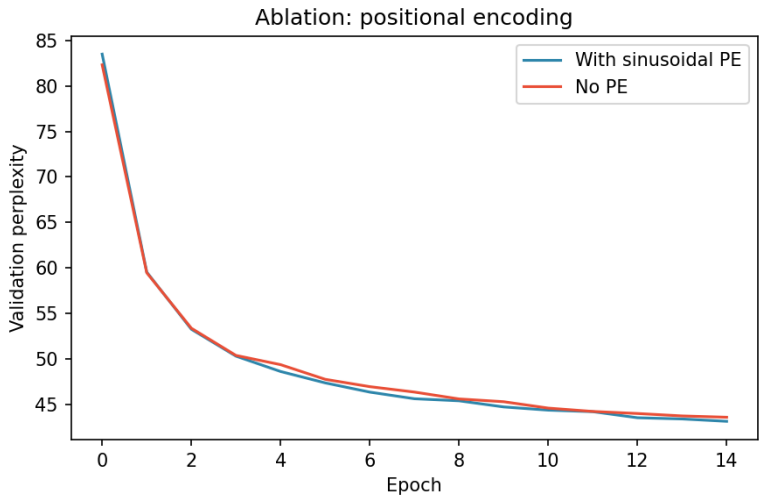

Figure 4: Positional encoding ablation: validation PPL for epochs 1–15, main model (truncated) vs. no-PE model.

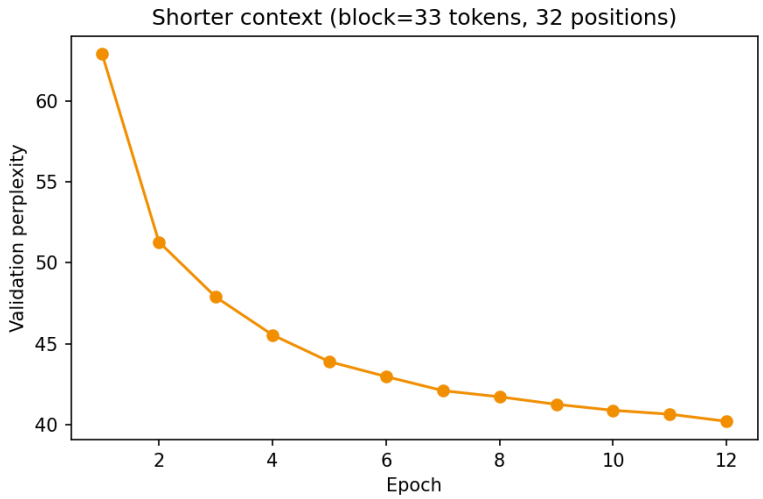

Figure 5: Shorter context (block 33, stride 16): validation PPL over 12 epochs.

3.4 Qualitative generation (optional)

We sampled text from the trained model with temperature ≈ 0.85. Output is Shakespeare-*flavored* but not coherent prose, e.g. fragmented lines such as "That is, a woman; if you be more, you'er's thereock," mixed with stage-like labels. This matches expectations for a tiny model and limited training budget.

4 Discussion

Stability. In development, the most sensitive knobs were learning rate and gradient clipping; without clipping, loss occasionally spiked. Batch size 32 and AdamW defaults proved stable for 25 epochs.

Positional information. The ablation gives concrete evidence that sinusoidal PE improves validation PPL relative to an otherwise identical stack, consistent with the idea that self-attention is permutation-invariant in the absence of position tags.

Memory and runtime. For batch B, heads H, and sequence length T, attention materializes $O(BHT^2)$ scores (before optimizations such as FlashAttention). Training therefore favors modest T on consumer hardware; the short-context experiment is a deliberate nod to that constraint.

Course context. This assignment uses **supervised** next-token CE. It differs from **RLHF/PPO** (L11), where a policy is tuned against a learned reward; here the objective is fully specified by human text labels.

5 AI Tool Usage Disclosure

Chat-based AI tools assisted with **initial** notebook scaffolding, plotting layout, and **first-draft** wording of this report. I **ran all experiments** (training, ablations, figures, and optional generation) in the submitted notebook, **verified** numerical outputs and plot filenames, **edited** the report for accuracy (including epoch-matched ablation wording and non-comparability of the short-context run), and **interpreted** attention maps and metrics. Figures in the PDF match the notebook's saved `fig_*.png` files.